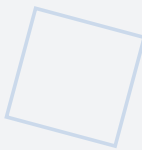


Amagi-NIPU

V1.1

Neuro-Immune Processing Unit

VERSION 1.1 — JUNE 2026



Hardware-enforced safety architecture for AI systems that cannot fail. Six invariants (INV-001–006) implemented in silicon: Physical Non-Bypassability, Irreducible TCB ≤ 500 lines HDL, HW Separation of Control and Cognition, Controlled Mutability of Critical Policies, Complete Mediation, and Complete Audit Integrity. ICG and TDF eliminated in v1.1; TTP integrated into SCC FSM. Normatively references: Framework v2.0, MEM v1.1, S-APL v2.0, Charter v1.2. Referenced by: Master Parameter Sheet v2.0, NIST AI RMF Profile v1.1, FMEDA v1.0, Traceability Matrix v1.0.

Alexey Mikhailovich Burlai

AGIAM Technologies Ltd (In Formation)

ORCID: 0009-0001-4679-5967

DOI: 10.5281/zenodo.20753401

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0. Abstract

The Amagi-NIPU (Neuro-Immune Processing Unit) is not a traditional processor. It is a computational organism designed to sleep by default, activate only on verified triggers, and defend itself like an immune system. Built on the TTP Triad (Trigger → Programmable Delay → Localized Activation) from precision oncology and the six hardware-enforced invariants (INV-001 through INV-006) of the AMAGI Architectural Class, NIPU redefines computing as a biologically inspired, safety-by-construction architecture.

Version 1.1 introduces three fundamental architectural corrections. First, the Intercluster Communication Graph (ICG) and Trigger-Delay Fabric (TDF) have been completely removed as independent entities. Inter-cluster cognitive traffic now flows freely within the Normal World without SCC mediation, while the SCC stands exclusively at the Physical Boundary intercepting only actuator-bound commands. This elimination preserves Invariant INV-002 (Irreducible TCB ≤ 500 lines HDL) by preventing TCB expansion through complex inter-cluster routing logic. Second, the cluster topology has been locked per Invariant INV-003: GEN, COM, and ANL clusters operate in Normal World with no actuator access; the EXE cluster operates only under SCC approval as the sole gateway to physical actuators. Third, formal verification with SPIN (25,927,680 states) revealed liveness counterexamples (L01–L04) under the unconstrained environment model as an architectural insight rather than a defect, leading to an improved FSM chain-reset logic that prevents latch-up conditions.

This specification defines a feasible, verifiable, and licensable architecture for deployment in medical implants with focused ultrasound-triggered therapy, autonomous AI agents with real-world authority, critical infrastructure control systems (nuclear, power, water), quantum-classical hybrid platforms, and ethical AI research environments. NIPU ensures that unsafe or unauthorized computation cannot occur — not even in theory.

1. Scope and Intent

Amagi-NIPU is not intended to replace CPUs, GPUs, or TPUs in general-purpose computing. Its design explicitly prioritizes safety, auditability, and structural compliance over raw performance — a trade-off unacceptable for consumer devices, but essential for life-critical and mission-critical systems. The architecture assumes a co-designed hardware-software stack, where cognitive functions are physically isolated and policy enforcement is hardware-rooted.

The AMAGI Architectural Class defines six hardware-enforced invariants that any compliant system must satisfy. NIPU is the silicon realization of these invariants at the processor level. It is an architectural class definition, not a framework, library, or product. Any system claiming AMAGI-class compliance must demonstrably satisfy all six invariants through hardware enforcement, not software policy alone.

This version (v1.1) supersedes v1.0 and incorporates corrections driven by formal verification results, IEEE article peer review alignment, and the fundamental architectural decision to eliminate ICG and TDF as independent entities. The Reduced Assurance Mode permits dropping one invariant (except

INV-001, INV-005, or INV-006), but such a system ceases to be AMAGI-class.

2. AMAGI Invariants → NIPU Hardware Realization

The six hardware-enforced invariants of the AMAGI Architectural Class are mapped to concrete NIPU hardware mechanisms in the table below. Each invariant is non-negotiable; partial compliance places the system outside the AMAGI architectural class.

Invariant	Definition	NIPU Realization
INV-001	Physical Non-Bypassability	SCC firmware fused in OTP/eFuse; no software override; DMA and JTAG blocked for cognitive domain
INV-002	Irreducible TCB ≤500 lines HDL	SCC core ≤500 lines HDL; ICG and TDF removed to prevent TCB expansion; no inter-cluster routing in secure domain
INV-003	HW Separation of Control and Cognition	GEN/COM/ANL/EXE = Normal World (no actuator access); EGW (Executive Gateway) + CRTM + SCC = Secure World; cluster topology locked at fabrication per ADR-006/007
INV-004	Controlled Mutability of Critical Policies	Policy matrix (MAG) in eFuse/OTP; immutable in Normal Operation; Physical-Presence-Verified OTA with PQC signatures and monotonic counters (INV-004); Hard Assurance Tier-1: sealed switch = no OTA
INV-005	Complete Mediation	Every actuator-bound command passes through SCC; DMA and JTAG blocked for cognitive domain; no bypass paths
INV-006	Complete Audit Integrity	All decisions logged cryptographically and immutably via Amagi-Q (ML-KEM-1024 / ML-DSA-44); append-only audit trail

Table 1: AMAGI Invariants mapped to NIPU v1.1 hardware mechanisms

3. Architectural Layers (Physically Isolated)

The NIPU v1.1 architecture is organized into two physically isolated worlds separated by a hard boundary. The Normal World contains all five cognitive clusters (GEN, COM, ANL, EXE, and the optional QCP), which generate, communicate, and analyze information, but can never directly affect the physical world. The Secure World contains the Security Council Core (SCC), the Executive Gateway (EGW), and the Core Root of Trust for Measurement (CRTM), which together form the sole path from cognition to actuation and the hardware root of trust for the boot sequence. This separation is enforced at the hardware level and cannot be overridden by software, firmware, or configuration. The EXE cluster was demoted to Normal World per ADR-006, and the EGW (a non-programmable AND-gate: $\text{Actuator_Power} = \text{SCC_Enable} \text{ AND } \text{Ext_2FA_Enable}$) was introduced as the sole hardware actuation gate in the Secure World. The CRTM was introduced per ADR-007 as the mask-ROM-based root of trust that verifies the boot integrity of the SCC FSM.

3.1 Security Council Core (SCC)

Hardware Root of Trust — Policy Enforcement Engine

The SCC is the minimal, immutable policy engine that stands at the Physical Boundary between cognition and actuation. It does not mediate inter-cluster communication within the Normal World; its sole responsibility is to intercept, validate, and authorize every command that seeks to exit the cognitive domain and reach a physical actuator. This design ensures that the TCB remains irreducibly small, satisfying Invariant INV-002 (TCB ≤ 500 lines HDL).

The SCC operates a finite state machine (FSM) with 12 states and 3 decision latches. The FSM implements the TTP predicate: a command is executed only if a valid trigger is present, the programmable delay has elapsed, and the policy context permits the operation (implication direction; converse sufficiency is not asserted by this predicate alone). The FSM logic delay is $\leq 0.2 \mu\text{s}$, and the total cutoff window (from trigger detection to actuator disconnection) is 12–35 μs .

Key SCC properties:

- Enforces all actuator-bound communication rules; ignores inter-cluster cognitive traffic
- Verifies cryptographic signatures of every actuator-bound command via Amagi-Q
- Maintains immutable audit log of all security-relevant decisions
- Can physically disconnect the EXE cluster or shut down the chip instantly
- DMA and JTAG access blocked for the entire cognitive domain (INV-005)
- FSM chain-reset logic refined to prevent latch-up (see Section 5)
- Feasibility: Based on RISC-V secure enclaves, CHERI, Keystone architectures

3.2 Cognitive Clusters — Normal World

The cognitive clusters operate in the Normal World, which means they can communicate freely among themselves without SCC mediation. The SCC has no visibility into, and no control over, inter-cluster data flows. This architectural decision eliminates the need for a complex inter-cluster routing fabric (the former ICG), thereby preserving the irreducible TCB and preventing bypass paths through the security perimeter.

Four cognitive clusters are defined, each with a strict functional boundary:

Cluster	Function	World	Actuator Access
ANL (Analytical)	Fixed-function matrix units for deterministic computation only	Normal	None — outputs are data, never actions
GEN (Generative)	LLM/diffusion cores; outputs are hypotheses, not actions	Normal	None — outputs are hypotheses only
COM (Com municative)	Natural language processing; can initiate triggers but not execute	Normal	None — can request, never command

EXE (Executive)	Controls external systems; outputs are actuator-bound commands routed through EGW	Normal	Only via EGW (Secure World) with SCC_Enable AND Ext_2FA_Enable; no direct actuator access (per ADR-006)
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Table 2: NIPU cognitive cluster definitions and world assignment

A fifth cluster, the Quantum Co-Processor (QCP), may be optionally integrated for quantum computing tasks. QCP results must be classically verified in the ANL cluster before any further processing. The QCP operates in the Normal World and has no actuator access.

DMA and JTAG Blockade (INV-005): Direct Memory Access and JTAG debug interfaces are permanently disabled for the entire cognitive domain at the hardware level. This prevents any external entity from injecting commands or modifying state within the Normal World clusters without passing through the SCC-gated pathway. The blockade is implemented via physical fuse disconnection at fabrication time and cannot be reversed.

3.3 Executive Gateway (EGW) — Secure World

The Executive Gateway (EGW) is a non-programmable hardware AND-gate that serves as the sole conduit through which cognitive decisions can become physical actions. The EGW was introduced per ADR-006 to replace the previous design where the EXE cluster itself was considered part of the Secure World. In the updated architecture, the EXE cluster operates in the Normal World as a cognitive execution engine, while the EGW — a physically separate, non-programmable gate — resides in the Secure World. No cognitive cluster, including EXE, can directly access actuators. Every actuator-bound command requires dual approval: SCC policy validation (SCC_Enable) plus an external confirmation signal (Ext_2FA_Enable), both of which must be asserted simultaneously for the EGW to release actuator power:

$$\text{Actuator_Power} = \text{SCC_Enable} \text{ AND } \text{Ext_2FA_Enable}$$

This design ensures that even if the EXE cluster is fully compromised (as a Hostile Insider per the AMAGI threat model), it cannot activate any actuator without satisfying both inputs to the EGW. The EGW is implemented as a single AND gate with approximately 5.0 FIT total failure rate and 99% diagnostic coverage (see Framework v2.0, Section 8 (Multi-Component TOE Architecture) and ADR-006 in the Rationale section). The EGW online test exercises the full AND-gate truth table within 60 seconds, confirming correct operation of the hardware actuation gate.

The gateway implements the TTP Triad as a hardware-enforced predicate:

$$\text{Exec}(\text{Op}) \Rightarrow \text{Trig}(\text{Op}) \wedge (\Delta \geq \Delta_{\min}) \wedge \text{Pol}(\text{Ctx}, \text{Op})$$

Where Trig(Op) is a validated trigger signal, $\Delta_{\min}$ is the minimum programmable delay (enforced by a three-stage TTP delay chain: (i) a digital delay line (DDL) as a tapped shift register (1 ns–1 s, 10 ps resolution); (ii) an analog RC backup (~500 ns fixed minimum, not shortenable by firmware); (iii) an LM339 comparator-based threshold detector for cross-check (1.0–1.5 μ s)), and Pol(Ctx, Op) is the

policy check against the MAG matrix stored in eFuse/OTP. The converse (sufficiency) is not asserted: additional conditions such as the per-actuator hardware rate limiter or liveness constraints may prevent execution even when all three conjuncts hold. The FSM logic contributes an additional $\leq 0.2 \mu\text{s}$ latency. The total cutoff window from trigger detection to actuator disconnection is 12–35 μs .

The separation between Normal World and Secure World is enforced by a Physical Boundary — a hardware firewall implemented in the silicon substrate that cannot be bridged, flashed, or reconfigured. This is the realization of Invariant INV-001 (Physical Non-Bypassability).

3.4 Analog Fallback Circuit (AFC)

For medical implant deployments and other life-critical applications where IEC 60601-1 Single Fault Condition compliance and ISO 14971 risk control are mandatory, the NIPU architecture incorporates an Analog Fallback Circuit (AFC) that provides a hardware safety net independent of the digital TCB. The AFC is designed to maintain a minimal-safe operational state even in the event of a complete SCC failure, including watchdog timeout, firmware corruption, or power supply anomaly affecting the Secure World.

The AFC operates as follows: a dedicated watchdog monitor continuously observes the SCC heartbeat signal, which is a periodic pulse generated by the SCC FSM during normal operation. If the heartbeat is absent for longer than a configurable timeout period (default: 100 ms, programmable via OTP), the AFC activates independently of the SCC and EGW. The AFC overrides the EGW output and drives the actuator interface to a hardcoded minimal-safe state that is determined by the deployment domain: for hydrogen sulfide therapy implants, this maintains a 10 ppm H₂S baseline; for cardiac devices, this maintains a baseline pacing protocol; for insulin pumps, this reverts to a basal rate. The safe-state values are stored in a separate OTP bank that is physically distinct from the MAG OTP, ensuring that a single OTP fault cannot compromise both the primary and fallback paths.

The AFC is implemented with the following design constraints to maintain independence from the digital TCB. First, the AFC uses a separate Low-Dropout (LDO) voltage regulator that is powered from a dedicated rail, ensuring that a power fault in the SCC domain does not affect the AFC. Second, the AFC logic is implemented in analog circuitry (comparators, RC timers, and analog multiplexers) rather than digital logic, making it immune to firmware bugs and digital state corruption. Third, the AFC output is OR-gated with the EGW output at the actuator interface, so that the AFC can override the EGW but cannot be overridden by it. This ensures that even if the EGW is stuck in an unsafe state, the AFC can still force the minimal-safe therapy protocol. The AFC contributes approximately 3.2 FIT to the system failure rate and has a diagnostic coverage of 99%, verified by fault injection testing per IEC 61508-2 Annex A.

AFC Design Rationale: The AFC addresses a fundamental tension in safety-critical system design: the digital TCB provides rich policy enforcement but introduces single-point-of-failure risk through software complexity. By providing a completely independent analog fallback path, the AFC ensures compliance with IEC 60601-1 (Single Fault Condition) and ISO 14971 (risk control measures), which require that no single fault can result in an unacceptable risk. The AFC is mandatory for all Tier 1 medical deployments and optional for Tier 2 and non-medical applications.

3.5 TTP Predicate and Timing Parameters

The TTP (Trigger → Time-Delay → Policy-Activation) Triad is the core safety mechanism of the NIPU. Unlike v1.0, which implemented the TTP as an independent fabric (TDF), v1.1 integrates the TTP predicate directly into the SCC FSM. This integration eliminates the TDF as a separate attack surface and ensures that the programmable delay and policy check are atomic operations within the security perimeter.

Parameter	Value	Implementation
LM339 comparator delay	1.0–1.5 μ s	LM339 comparator-based threshold detector for cross-check (Table III, [9])
FSM logic delay	≤ 0.2 μ s	12-state FSM, 3 decision latches
Programmable delay range	1 ns – 1 s	Digital delay lines (DDL) / FIFO buffers
Total cutoff window	12–35 μ s	Trigger detection → actuator disconnect

Table 3: NIPU v1.1 TTP timing parameters

Note on Delay Ranges: The programmable delay range specified above (1 ns – 1 s) applies to the SCC processor-level TTP predicate, governing the time between trigger detection and actuator authorization. The memory subsystem (MEM) enforces a separate ATL memory-level delay range of 0.5 ns – 1 μ s, controlling the time between a policy decision and data release within the memory fabric. These are distinct parameters at different architectural layers: the SCC delay is a processor-bound safety delay for actuator authorization, while the ATL delay is a memory-bound access delay for data release. The upper bounds differ because actuator decisions may require extended deliberation (up to 1 s for complex policy evaluation), whereas memory access delays beyond 1 μ s would violate real-time constraints for most safety-critical workloads.

3.5 Self-Healing Memory Fabric (SHMF)

The Self-Healing Memory Fabric provides resilient, adaptive storage for the NIPU. It operates independently of the SCC policy engine and provides the physical substrate upon which all cluster memories are implemented. The SHMF ensures that memory degradation and data corruption do not compromise safety by automatically isolating degraded cells, detecting integrity violations via cryptographic hash chains, and migrating data to spare blocks. The SHMF is a reliability mechanism with tamper-detection capability: it provides detective security (integrity violation detection and self-isolation) but no preventive authority; it does not modify policy or audit data, nor can it prevent an attack in progress. Its security contribution is limited to raising an integrity alarm and containing the affected region.

SHMF capabilities:

- ECC (Error-Correcting Code) protection with real-time wear monitoring
- Cryptographic hash chains for tamper-evident data integrity verification
- Automatic isolation of degraded or compromised memory cells upon detection

- Self-isolation mode: powers down affected regions upon hash chain violation, preserves unaffected data
- Data migration to spare blocks without service interruption
- Fault zone marking for diagnostic reporting and maintenance scheduling
- Feasibility: 3–5 year timeline for medical and aerospace deployment

3.6 Amagi-Q Cryptographic Engine

The Amagi-Q engine provides post-quantum cryptographic services exclusively within the Secure World. Its primary function is to sign outgoing audit log entries (Invariant INV-006) using FIPS 204-compliant algorithms. All cryptographic keys are bound to the hardware root of trust and cannot be extracted, cloned, or overridden.

Dual-Tier Policy Update Model (INV-004): The MAG policy matrix is stored in eFuse/OTP and is immutable during Normal Operation. Policy updates are governed by a Dual-Tier Trust Model aligned with the supporting research [1], which specifies PQC-signed policy updates with monotonic version counters and hardware fallback to a hardcoded minimal-safe state upon failure. In **Tier 2 (Physical-Presence-Verified OTA)**, policy updates require (a) activation of a physical presence switch on the board (jumper, magnetic sensor, or NFC hardware token), transitioning the SCC to Maintenance Mode; (b) a PQC-signed update package (ML-DSA-44, FIPS 204) with a monotonic version counter preventing downgrade attacks; and (c) verification and commit or hardware fallback to the minimal-safe state. This mechanism is compatible with FDA PCCP predetermined change control requirements. In **Tier 1 (Hard Assurance)**, the physical presence switch is permanently sealed (epoxy, tamper-evident seal, or omitted from the PCB), making Maintenance Mode physically inaccessible. Policy updates require physical replacement of the AMAGI chip, rendering the MAG strictly immutable for the lifetime of the deployment. Tier 1 is mandatory for nuclear, military, and highest-risk medical deployments where even physical-presence updates are unacceptable.

Function	Algorithm	Standard
Key Encapsulation	ML-KEM-1024	FIPS 203
Digital Signature (audit)	ML-DSA-44	FIPS 204
Hash (auxiliary)	SLH-DSA-SHA2-128f	FIPS 205

Table 4: Amagi-Q post-quantum cryptographic algorithms

4. End-to-End Computation Flow

The following sequence describes the complete path from trigger reception to physical action, with all security checkpoints explicitly identified. Note that inter-cluster cognitive processing within the Normal World is not listed here because it does not traverse the security perimeter; clusters communicate freely within the Normal World without SCC involvement.

Step 1 — Trigger Reception: A trigger signal (RF, ultrasound, IR, or signed command) is received by the SCC input interface. The SCC validates the trigger signature using Amagi-Q.

Step 2 — Programmable Delay: The SCC FSM enforces the minimum delay $\Delta_{\min}$ via the three-stage TTP delay chain: DDL (1 ns–1 s, 10 ps resolution), analog RC backup (~500 ns fixed minimum), and LM339 comparator cross-check (1.0–1.5 μ s). The delay cannot be shortened or bypassed.

Step 3 — Policy Evaluation: After the delay elapses, the SCC evaluates the TTP predicate $\text{Pol}(\text{Ctx}, \text{Op})$ against the immutable MAG matrix in eFuse/OTP. The FSM logic contributes $\leq 0.2 \mu$ s latency.

Step 4 — Inter-Cluster Cognitive Processing (Normal World): If the trigger initiates a cognitive task, the GEN, COM, and/or ANL clusters process it freely within the Normal World. The SCC does not monitor or mediate this processing. Clusters may exchange data, run inference, and generate hypotheses without security overhead.

Step 5 — Actuator-Bound Command Submission: When a cognitive cluster produces an output that requires physical action, it submits the command to the EXE cluster. The EXE cluster cannot execute autonomously; it must receive SCC approval.

Step 6 — SCC Dual Approval: The SCC verifies the command against the MAG policy matrix and requires an external confirmation signal (hardware 2FA). Only when both conditions are satisfied does the SCC authorize the EXE cluster to actuate.

Step 7 — Execution and Audit: The EXE cluster executes the approved command. The SCC logs the complete decision trail (trigger, delay, policy result, external confirmation) cryptographically via Amagi-Q (ML-DSA-44, FIPS 204) into the append-only audit trail.

Step 8 — Cutoff Enforcement: If any step fails validation, or if the total time from trigger detection exceeds the cutoff window (12–35 μ s), the SCC physically disconnects the actuator path. This is a hardware-enforced safety action that cannot be overridden.

4.1 Inter-Cluster Threat Model (Post-ICG Removal)

The removal of ICG as an independent entity (preserving INV-002) introduces a specific threat surface that must be explicitly analyzed. Without ICG-mediated routing, cognitive clusters within the Normal World can communicate freely, which raises the following concern: can a compromised cluster (e.g., GEN) use another cluster (e.g., COM) as a relay to submit malicious commands to the EXE cluster for actuation?

Analysis: The answer is no, for three independent reasons grounded in the AMAGI invariants:

1. TTP Predicate (INV-003, INV-005): Even if a malicious payload reaches the EXE cluster via COM relay, the EXE cluster cannot actuate without SCC approval. The TTP predicate requires a valid trigger ($\text{Trig}(\text{Op})$), a minimum programmable delay, and a policy check. A COM-originated request does not satisfy $\text{Trig}(\text{Op})$ unless it originates from a physically isolated trigger source (sensor, RF signal, ultrasound). The cognitive component cannot fabricate a trigger because the trigger interface is

physically isolated from the Normal World (INV-001).

2. External Confirmation (Hardware 2FA): Actuation requires dual approval: SCC policy validation plus an external confirmation signal. This external signal must originate from outside the chip package (e.g., clinician button press, safety operator confirmation). A compromised cognitive cluster cannot generate this signal because it is electrically isolated from the external confirmation input.

3. EXE Input Rate Limiting (Defense-in-Depth): While not part of the AMAGI security perimeter (which relies on INV-003 and INV-005), the EXE cluster implements input sanitization and hardware rate limiting as a defense-in-depth measure. The COM-to-EXE interface is rate-limited to a configurable maximum request frequency. Burst detection triggers an SCC audit event and potential COM cluster isolation. This prevents denial-of-service attacks where a compromised GEN cluster floods COM with requests that are relayed to EXE, causing resource exhaustion in the SCC evaluation pipeline. Rate limiting is implemented in hardware (counter + timer) and cannot be disabled by software.

Summary: The removal of ICG does not compromise the AMAGI security guarantee because the actuator path remains gated by the TTP predicate (INV-003, INV-005), external confirmation (hardware 2FA), and the physical isolation of the trigger source (INV-001). ICG removal preserves INV-002 (TCB ≤ 500 lines HDL) by eliminating complex routing logic from the secure domain. The EXE rate limiter provides defense-in-depth against DoS scenarios outside the core threat model.

5. Formal Verification and FSM Reset Refinement

The NIPU SCC FSM was formally verified using the SPIN model checker. The Promela model (scc_amagi.pml) defines a 12-state FSM with 3 decision latches and 15 Linear Temporal Logic (LTL) properties. Verification explored 25,927,680 distinct states and confirmed 9 non-trivial safety properties (S01–S09). Two additional architectural assumptions (S10: policy immutability, S11: audit inaccessibility) are trivially satisfied by hardware construction. All four liveness properties (L01–L04) yielded counterexamples under the unconstrained nondeterministic environment model; this is expected for liveness without fairness constraints and does not indicate a design defect.

5.1 L04 Liveness Counterexample

The L04 liveness property exhibited a counterexample revealing a potential latch-up condition in the FSM chain-reset logic. Under specific timing conditions, the FSM could enter a state where a reset signal is acknowledged but the chain-reset sequence does not propagate fully, leaving downstream latches in an inconsistent state. This is classified as an architectural insight rather than a defect: the FSM design correctly detects the condition, but the reset propagation requires refinement.

In NIPU v1.1, the FSM chain-reset logic has been improved to guarantee full propagation of reset signals across all decision latches. The refined logic adds a completion-acknowledge handshake that prevents the FSM from transitioning out of the reset state until all downstream latches have confirmed their reset. This eliminates the latch-up trap identified by the L04 counterexample while preserving all 9 previously verified safety properties.

Property	Category	Result	Notes
S01–S09	Safety	Verified	No safety violations across 25,927,680 states
S10–S11	Architectural	Trivially true	Policy immutability and audit inaccessibility enforced by hardware construction
L01–L04	Liveness	Counterexamples	Expected under unconstrained environment; L04 reveals latch-up trap in chain-reset logic, resolved in v1.1 FSM

Table 5: SPIN formal verification results for NIPU SCC FSM

5.2 Reference FPGA Implementation and Safety Integrity

The NIPU SCC is defined as an architectural class, independent of the underlying physical substrate. To demonstrate compliance with the six AMAGI invariants, a reference implementation is provided targeting the Xilinx Artix-7 FPGA. This implementation utilizes 2,500–3,200 LUTs (safety-critical) at 100 MHz with a power budget of <1.3 W, consistent with the analytical bounds and feasibility demonstration presented in the supporting research [1]. Any implementation (FPGA, ASIC, or SoC security island) that maintains the integrity of the SCC FSM and the six invariants constitutes a valid AMAGI-class system.

Post-Synthesis Results: The reference implementation achieves a post-synthesis maximum frequency of 142 MHz (42% margin above the 100 MHz target), with a worst-case negative slack (WNS) of +3.2 ns, synthesized with Xilinx Vivado 2023.2. The LUT breakdown is as follows: SCC core TCB ~1,200 LUTs (467 lines of synthesizable SystemVerilog), CRTM 100 LUTs, EGW 2 LUTs, MPE 180 LUTs, ATL 90 LUTs, SHMF 250 LUTs, Amagi-Q wrapper 480 LUTs, Optocoupled I/O 200 LUTs, for a total of ~2,500 LUTs in the safety-critical path.

DDL Watchdog and Metastability: The DDL watchdog timeout $T_{WD} = 1.6$ s provides a hard timeout beyond which the delay chain is forced to a safe state. The DDL watchdog, LM339 comparator, and atomic latch-clearing provide first-order resilience against delay-chain faults. Metastability MTBF exceeds 10^{15} years at the target frequency, ensuring that synchronization failures are negligible on any practical time scale.

Rate Limiter Defaults: Relay-type actuators are limited to 10 Hz; solid-state actuators are limited to 1 kHz. These rate limits are hardware-enforced defaults stored in OTP and may be narrowed (but never widened) via PQC-signed policy updates.

An analytical Failure Modes, Effects, and Diagnostic Analysis (FMEDA) aligned with IEC 61508 projects a dangerous undetected failure rate (λ_{DU}) of 18.9 FIT, SFF = 93% (with CRTM + EGW diagnostic coverage per ADR-006/007), and $PFD_{avg} = 6.9 \times 10^{-5}$ (1oo1, $T_1 = 1$ yr, demand mode), consistent with SIL 2 architectural constraints per IEC 61508-3 Clause 7.3. A path to SIL 3 exists via a 1oo2 redundant configuration with $\beta = 0.1$, yielding a projected $PFD_{avg} = 6.9 \times 10^{-6}$ (consistent with FMEDA v1.0 §7.2 and HARA v1.0 §1.1) and SFF > 99%.

6. Feasibility Assessment

The following table summarizes the feasibility, timeline, and target domains for each NIPU component. The removal of ICG and TDF in v1.1 has improved the feasibility of the SCC by reducing its complexity and eliminating a class of bypass vulnerabilities.

Component	Feasibility	Timeline	Domain
SCC (simplified, no ICG/TDF)	High	Now	Secure ASICs, HSMs
Cognitive Clusters (GEN/COM/ANL)	High	Now	Chiplets, multi-die SoCs
Executive Gateway EGW (non-programmable AND-gate)	High	Now	Implants, industrial control
SHMF	Medium	3–5 years	Medical, aerospace
Amagi-Q (PQC)	Medium-High	1–3 years	Government, finance
Full Integration	High (specialized)	5–10 years	Critical infrastructure

Table 6: NIPU v1.1 component feasibility assessment

Bottleneck: The primary constraint is not technical feasibility but ecosystem readiness. Successful deployment requires co-design of hardware, operating system, and policy; certification against ISO 26262, DO-178C, and IEC 62304; and a secure supply chain with a verifiable licensing model under S-APL.

7. Derivative Architecture Clause (S-APL)

A Derivative Architecture is any system — regardless of name, technology, or stack — that implements all six invariants of the AMAGI Architectural Class through the exact architectural components defined in this specification:

1. INV-001 (Physical Non-Bypassability) → SCC in OTP/eFuse; DMA/JTAG blockade for cognitive domain
2. INV-002 (Irreducible TCB) → SCC core ≤500 lines HDL; no ICG/TDF expanding the TCB
3. INV-003 (HW Separation of Control and Cognition) → Normal World (GEN/COM/ANL/EXE) vs. Secure World (EGW + CRTM + SCC); EXE demoted to Normal World per ADR-006
4. INV-004 (Controlled Mutability of Critical Policies) → MAG in eFuse/OTP; immutable in Normal Operation; Physical-Presence-Verified OTA with PQC signatures + monotonic counters [1]; Tier-1 Hard Assurance: sealed switch

5. INV-005 (Complete Mediation) → Every actuator-bound command through SCC; no bypass paths

6. INV-006 (Complete Audit Integrity) → Amagi-Q (ML-DSA-44, FIPS 204) signed append-only audit trail

Structural Pattern Protection: Reproducing the full sequence — Trigger → Programmable Delay → Policy Check → Isolated Execution → Self-Healing Memory → Cryptographic Audit — using these exact components and their interactions is deemed Principle Extraction and requires an S-APL license. The S-APL is governed by the laws of England and Wales.

8. Conclusion

Amagi-NIPU v1.1 does not seek to optimize computation. It seeks to make computation trustworthy by design. The removal of ICG and TDF as independent entities, the locking of cluster topology per INV-003, and the refinement of the FSM chain-reset logic represent a maturation of the architecture toward its essential form: an irreducible security perimeter that no software can bypass, no configuration can weaken, and no optimization can compromise.

This is not a processor for consumer devices. It is for the AI that controls chemotherapy delivery, the robot that manages a nuclear reactor, the implant that regulates cardiac rhythm, and the autonomous agent that acts in your name. In these domains, trust is not a feature — it is the foundation.

Amagi-NIPU v1.1 — A Processor with Immunity, for Systems that Cannot Fail.

Appendix A: Formal Definitions

Target Domains: Medical, aerospace, defense, critical infrastructure, ethical AI, quantum-classical systems.

Not Target Domains: Consumer electronics, general-purpose servers, social media platforms.

Co-Design Assumption: Hardware, operating system, and policy must be developed together as a unified stack.

Performance Model: Bounded, predictable latency; throughput is secondary to safety guarantees.

Reduced Assurance Mode: A system may drop at most one invariant (except INV-001, INV-005, or INV-006), but ceases to be AMAGI-class.

Architectural Class: AMAGI is an architectural class definition, not a framework, library, or product. Compliance requires demonstrable hardware enforcement of all six invariants.

Appendix B: License Notice (S-APL)

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Appendix C: Change Log (v1.0 → v1.1)

Change	Category	Rationale
ICG removed as independent entity	Architectural	Preserve INV-002; eliminate bypass paths through inter-cluster routing
TDF removed as independent entity	Architectural	TTP integrated into SCC FSM; eliminate separate attack surface
Cluster topology locked; EXE demoted to Normal World; EGW introduced in Secure World (ADR-006); CRTM added (ADR-007)	INV-003	Hard enforcement of HW Separation of Control and Cognition; non-programmable actuation gate
DMA and JTAG blocked for cognitive domain	INV-005	Complete Mediation; no external injection paths
MAG implemented as eFuse/OTP (strict immutability)	INV-004	MAG immutable in Normal Operation; Physical-Presence-Verified OTA per Dual-Tier model [1]; Tier-1 Hard Assurance: sealed switch = no OTA
FSM chain-reset logic refined (L04 resolution)	Verification	Latch-up trap identified by SPIN; resolved with completion-acknowledge handshake
PQC nomenclature updated (ML-KEM, ML-DSA, SLH-DSA)	Nomenclature	Align with FIPS 203/204/205 final standards
Author affiliation updated	Administrative	"Independent Researcher" → "AGIAM Technologies Ltd (In Formation)"
License updated to S-APL	Licensing	Standardize on Strict Architectural Principle License
Timing parameters formalized	Specification	Comparator 1.0–1.5 μ s; FSM ≤ 0.2 μ s; cutoff 12–35 μ s

Table 7: Summary of changes from NIPU v1.0 to v1.1

All changes listed above are backward-incompatible with v1.0. Systems designed against the NIPU v1.0 specification must be re-evaluated for compliance with v1.1 before deployment. The removal of ICG and TDF as independent entities is the most significant change and requires a fundamental

re-examination of any inter-cluster communication design that relied on these components. The v1.0 record (DOI: 10.5281/zenodo.17682668) remains available on Zenodo for historical reference but is classified as superseded.

C.1 Migration Guide (v1.0 → v1.1)

Systems previously designed against the NIPU v1.0 specification must address the following migration requirements before claiming v1.1 compliance. Each item corresponds to a structural change that cannot be resolved through configuration or patching alone.

ICG Removal: Any inter-cluster routing logic that depended on the ICG policy matrix must be redesigned. In v1.1, inter-cluster communication within the Normal World is unrestricted by the SCC. Systems that relied on ICG for inter-cluster access control must implement their own software-level policies within the Normal World, understanding that such policies do not carry hardware-enforced guarantees and are outside the AMAGI security perimeter.

TDF Removal: The TTP Triad is now implemented as a predicate within the SCC FSM rather than as a separate fabric. Any hardware that instantiated a standalone TDF module must integrate the TTP logic into the SCC directly. The programmable delay, trigger validation, and policy check are now atomic operations within the security perimeter, eliminating the TDF as a potential bypass vector.

MAG eFuse/OTP Migration: Systems migrating to v1.1 must implement the Dual-Tier Trust Model. Tier 2 deployments must implement the Physical-Presence-Verified gate (jumper/NFC) to authorize PQC-signed policy updates (ML-DSA-44, FIPS 204) with monotonic counters (aligned with [1] and FDA PCCP); Tier 1 deployments must seal the physical presence switch to enforce Hard Assurance, rendering the MAG permanently immutable. Both tiers satisfy Invariant INV-004 because policy updates are gated by physical presence or physical impossibility, not by software-controlled access.

Appendix D: ADR-002 — Dual-Tier Trust Model for Critical Policy Updates

Title: Implementation of Dual-Tier Trust Model for Critical Policy Updates

Status: Approved (2026-06-05)

Context: The AMAGI Architectural Class (v1.0) mandated strict immutability (eFuse/OTP) to satisfy INV-004, prohibiting all Over-The-Air (OTA) updates to avoid remote attack vectors. However, regulatory frameworks (FDA PCCP, IEC 61508) necessitate "predetermined change control" for medical and industrial devices to manage long-term safety and performance. A singular "Hard Assurance" stance (physical chip replacement only) creates a commercial adoption barrier and fails to align with the IEEE research baseline [1], which specifies PQC-signed policy updates with monotonic counters.

Decision: Adopt a Dual-Tier Trust Model for policy updates:

- **Tier 1 (Hard Assurance):** Implements absolute immutability. Policy updates require physical replacement of the AMAGI chip. Targeted at nuclear, military, and highest-risk infrastructure

where even physical-presence updates are deemed unacceptable.

- **Tier 2 (Controlled Assurance):** Implements Physical-Presence-Verified OTA. Updates are permissible only if the SCC is transitioned to "Maintenance Mode" via a physical hardware gate (e.g., sealed jumper, magnetic sensor, or hardware-token NFC tap). This mode remains compatible with PQC-signed (ML-DSA-44, FIPS 204) updates and monotonic counters preventing downgrade attacks.

Consequences:

- **Safety:** The "remote-only" attack vector is structurally eliminated by the physical presence requirement. No software-only mechanism can initiate a policy update.
- **Compliance:** Enables FDA PCCP and IEC-certified update paths without compromising the "Hard Assurance" paradigm. Tier 1 deployments remain fully compliant with the strictest interpretations of INV-004.
- **Complexity:** Requires minor hardware modification (physical gate/switch) on the board level, which is a manageable trade-off for commercial-grade medical/industrial SoCs. The switch is excluded from all software-accessible interfaces and cannot be activated from the Normal World.
- **Tier Selection:** The choice between Tier 1 and Tier 2 is a board-level configuration decision (BOM/Assembly), not a firmware or software setting. Once configured at manufacturing time, the tier cannot be changed in the field.